

❖ Networking Fundamentals and CIDR Basics

❖ What is CIDR? (Classless Inter-Domain Routing)

- **CIDR** stands for **Classless Inter-Domain Routing**. It's a method for allocating IP addresses and routing Internet Protocol packets that CIDR replaces the old **class-based IP addressing system** (Class A, B, C) by allowing **more flexible IP address allocation**.

❖ What is IP is an address?

- An **IP address** (Internet Protocol address) is a unique identifier assigned to a device connected to a network. It allows devices to **communicate with each other** over the Internet or local networks.
- IPv4 Address: 32-bit address divided into four 8-bit octets (e.g., 192.168.1.1)
- Written in dotted decimal format (e.g., 192.168.0.1)
- Each octet ranges from 0 to 255

❖ What is a Subnet Mask?

- A Subnet Mask determines which portion of the IP address refers to the network and which part refers to the host.
- It masks the IP address to identify the network boundary.

Example:

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- **This means:**
- ✓ First 24 bits (255.255.255) → Network portion
- ✓ Last 8 bits (0) → Host portion

❖ Classes of IP:

- Class A /8 | 0-126 (127.0.0.0 - Localhost / loopback address) | N.H.H.H | 0-255.0-255.0-255 | $256 \times 256 \times 256$ | 1C
- Class B /16 | 128 - 191 | N.N.H.H | 0-255.0-255 | 256×256 | 65536
- Class C /24 | 192 - 223 | N.N.N.H | 0-255 | 256
- Class D | 224 - 239 | Scitific and experimental use
- Class E | 240 - 255 | Future Use

❖ Relationship Between IP, Subnet Mask, and Network

- **IP Address** :- Identifies a **device (host)** on a network. **Example:** 192.168.1.10

- **Subnet Mask** :- Separates the IP into **network** and **host** parts. **Example**: 255.255.255.0
- **Network** :-A group of devices sharing the same network portion of the IP address.